

EDA & Predictive Modeling Requirements

- Develop an **EDA narrative** with **3–5 key findings**, supported by relevant charts and Provide a **predictive model overview**, including: Method, Input variables, Output results, Accuracy metric, **assumptions** and **Power BI dashboard wireframe**.

EDA Outcomes

- Defined the **data model and level of granularity**
- Identified key **dimensions** for BI and predictive analysis
- Identified relevant **measures**

Assumptions

- Other influencing (spurious) factors such as **policy changes** or **economic shifts** are assumed to remain constant during trend analysis.

Identified Measures

- Total Emissions** - Sum of greenhouse gas emissions across all provinces, sectors, and years. **NS Emissions** - Total emissions specific to Nova Scotia, used for focused regional analysis.
- NS GHG Change (%)** - Percentage change in Nova Scotia emissions between the earliest and latest years, indicating overall trend performance.
- Province Rank** - Ranking of provinces based on total emissions, used to compare relative emission levels across regions.

Table 1: Greenhouse Gas Emissions by Province and Sector (2013-2015)						
Year	Province	Sum of Emissions	Sector	Prov Highest Rank	NS Total Emissions	NS GHG Change %
2013	Alberta	257	Accommodation and food services	3	59	0.00
2013	Alberta	323	Administrative and support services	3	34	0.00
2013	Alberta	0	Advertising, promotion, meals, entertainment, and travel	1	0	0.00
2013	Alberta	13	Advertising, public relations and related services	3	1	0.00
2013	Alberta	6	Aerospace product and parts manufacturing	6	8	0.00
2013	Alberta	3065	Air transportation	3	381	0.00
2013	Alberta	47	Animal food manufacturing	3	8	0.00
2013	Alberta	22	Arts, entertainment and recreation	4	2	0.00
2013	Alberta	-1940	Balancing item: Aviation	10	-241	0.00
2013	Alberta	-9247	Balancing item: Biomass	10	-1528	0.00
2013	Alberta	-2268	Balancing item: Motor fuels	12	-250	0.00
2013	Alberta	4025	Balancing item: Non-energy Products from Fuels and Solvent Use	1	103	0.00
2013	Alberta	-5177	Balancing item: Other differences	13	671	0.00
2013	Alberta	1297	Balancing item: Synthetic fluorinated gases	4	270	0.00
2013	Alberta	4015	Balancing item: Waste	2	510	0.00
Total		7965876		1	170719	-0.25

Year

☐ 2013

☐ 2014

☐ 2015

Province

☐ Alberta

☐ British Columbia

☐ Manitoba

Sector

☐ Accommodation and food services

☐ Administrative and support services

☐ Advertising, promotion, meals, entertainment, and travel

☐ Advertising, public relations and related services

☐ Aerospace product and parts manufacturing

Year

- ☐ 2013
- ☐ 2014
- ☐ 2015
- ☐ 2016
- ☐ 2017
- ☐ 2018
- ☐ 2019
- ☐ 2020
- ☐ 2021
- ☐ 2022
- ☐ 2023

Province

- ☐ Alberta
- ☐ British Columbia
- ☐ Manitoba
- ☐ New Brunswick
- ☐ Newfoundland and Labrador
- ☐ Northwest Territories
- ☐ Nova Scotia
- ☐ Nunavut
- ☐ Ontario
- ☐ Prince Edward Island
- ☐ Quebec
- ☐ Saskatchewan
- ☐ Yukon

Sector

- ☐ Accommodation and food services
- ☐ Administrative and support services
- ☐ Advertising, promotion, meals, enterta...
- ☐ Advertising, public relations and relat...
- ☐ Aerospace product and parts manufac...
- ☐ Air transportation
- ☐ Animal food manufacturing
- ☐ Arts, entertainment and recreation
- ☐ Balancing item: Aviation
- ☐ Balancing item: Biomass
- ☐ Balancing item: Motor fuels
- ☐ Balancing item: Non-energy Products ...
- ☐ Balancing item: Other differences
- ☐ Balancing item: Synthetic fluorinated ...
- ☐ Balancing item: Waste

GHG Emissions Trend & Forecast — Nova Scotia

These predictions are based on existing GHG Emission trends to project future emissions trends based on historical data, using Power BI.

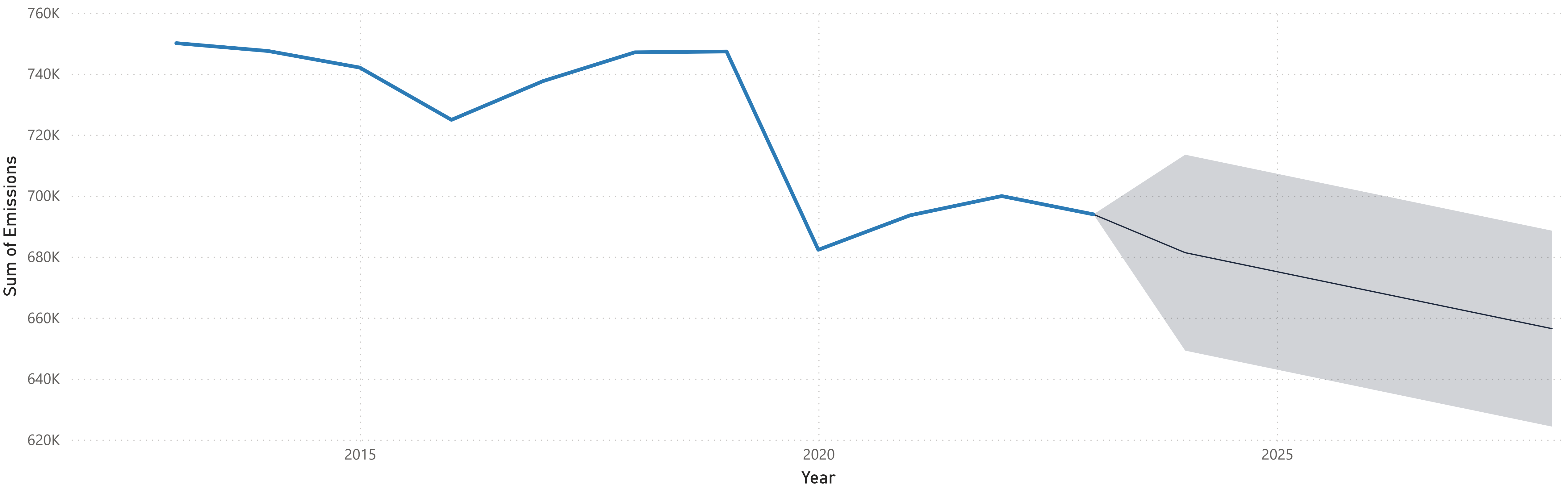
The forecast suggests that emissions are expected to continue declining if current patterns persist projects for the next five years.

The trend line shows a general downward direction, supporting the forecasted decline in emissions.

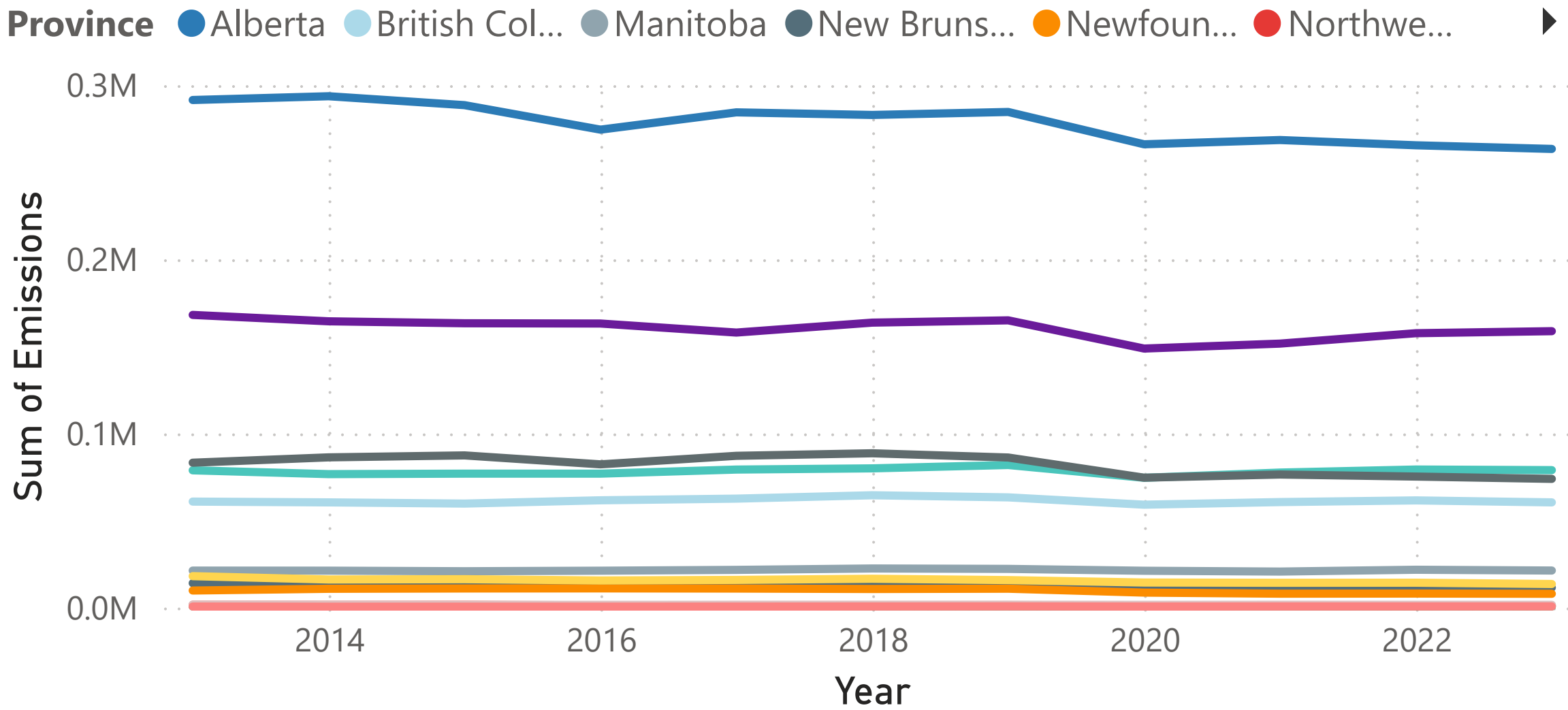
The forecast is based on historical trends and does not account for external factors such as policy changes or economic shifts, so the forecast should be viewed as a directional estimate rather than an exact prediction.

Overall, this forecast extends insights beyond historical data to support forward-looking environmental decision-making.

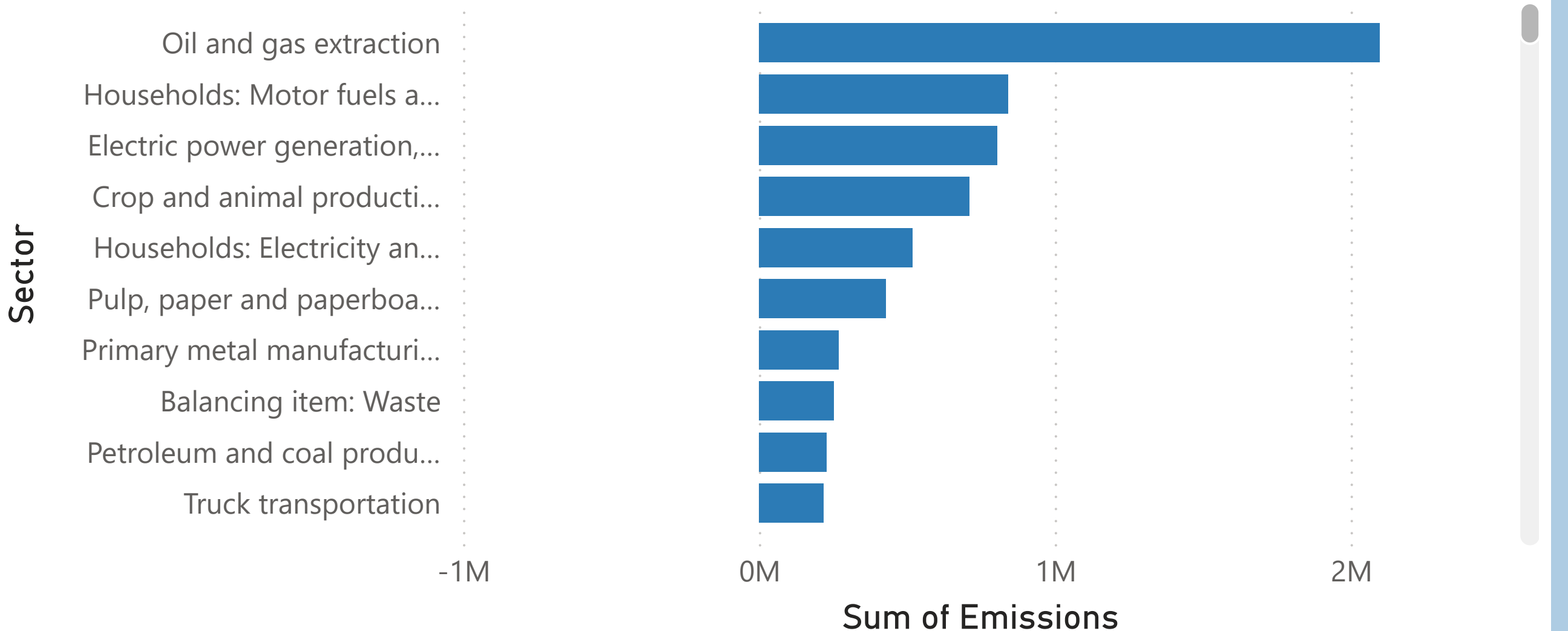
Sum of Emissions by Year



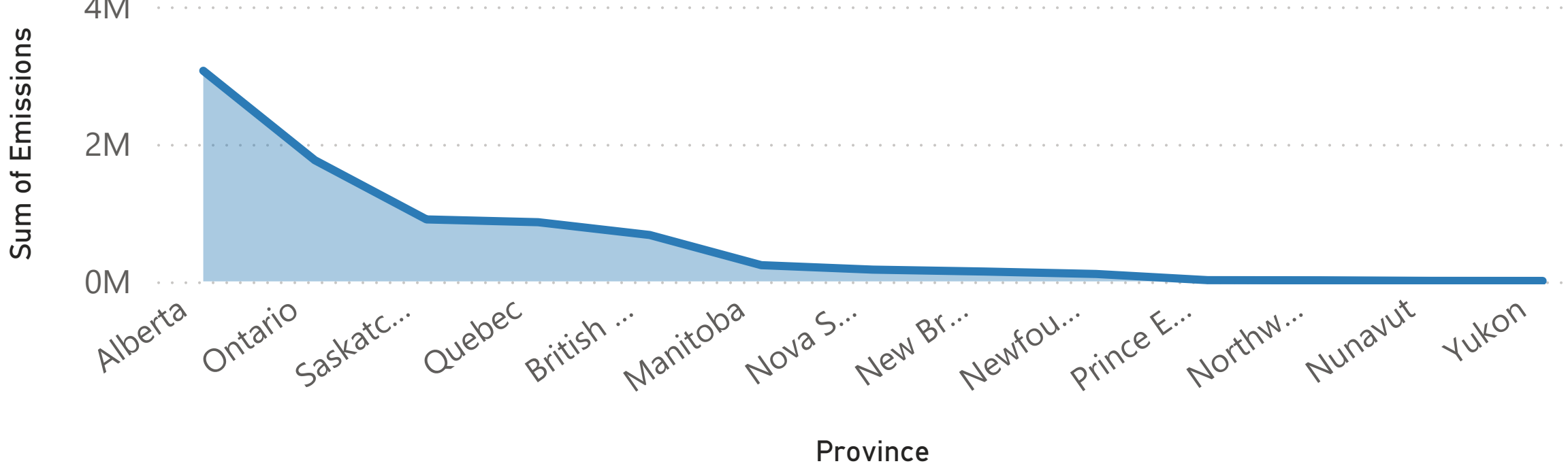
Sum of Emissions by Year and Province



Sum of Emissions by Sector



Sum of Emissions by Province



Year

2013	2017	2021
2014	2018	2022
2015	2019	2023
2016	2020	

Sector

☐	Accommodation and food services
☐	Administrative and support services
☐	Advertising, promotion, meals, entertainment, and telecommunications
☐	Advertising, public relations and related services
☐	Aerospace product and parts manufacturing
☐	Air transportation
☐	Animal food manufacturing

1

Prov Highest Rank

Province

Alberta	Manitoba	Newfoundland and Labrador	Nova Scotia	Ontario	Quebec	Yukon
British Columbia	New Brunswick	Northwest Territories	Nunavut	Prince Edward Island	Saskatchewan	

8M

Total Emissions Province

Year Slcr

2013

2014

2015

2016

2017

2018

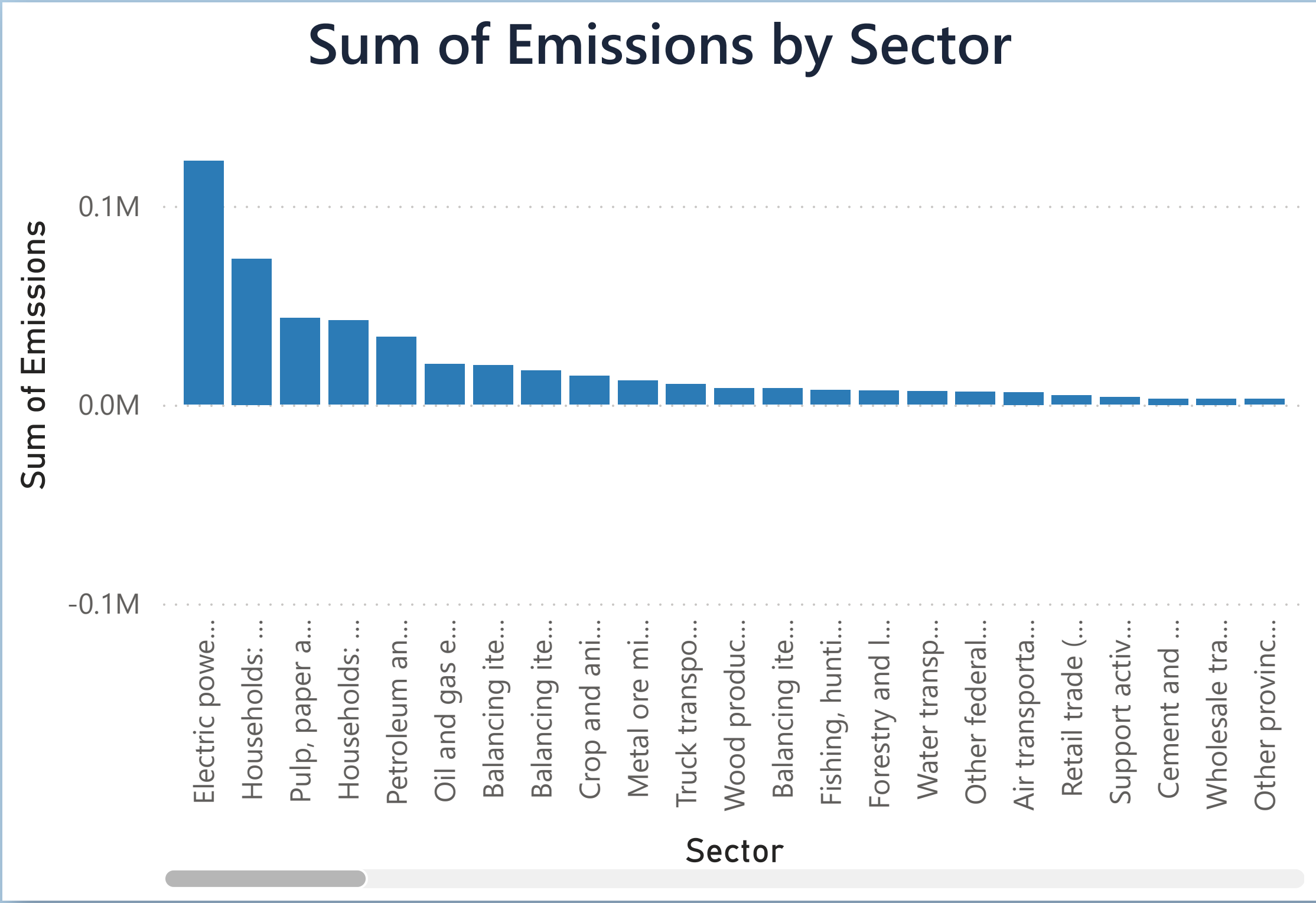
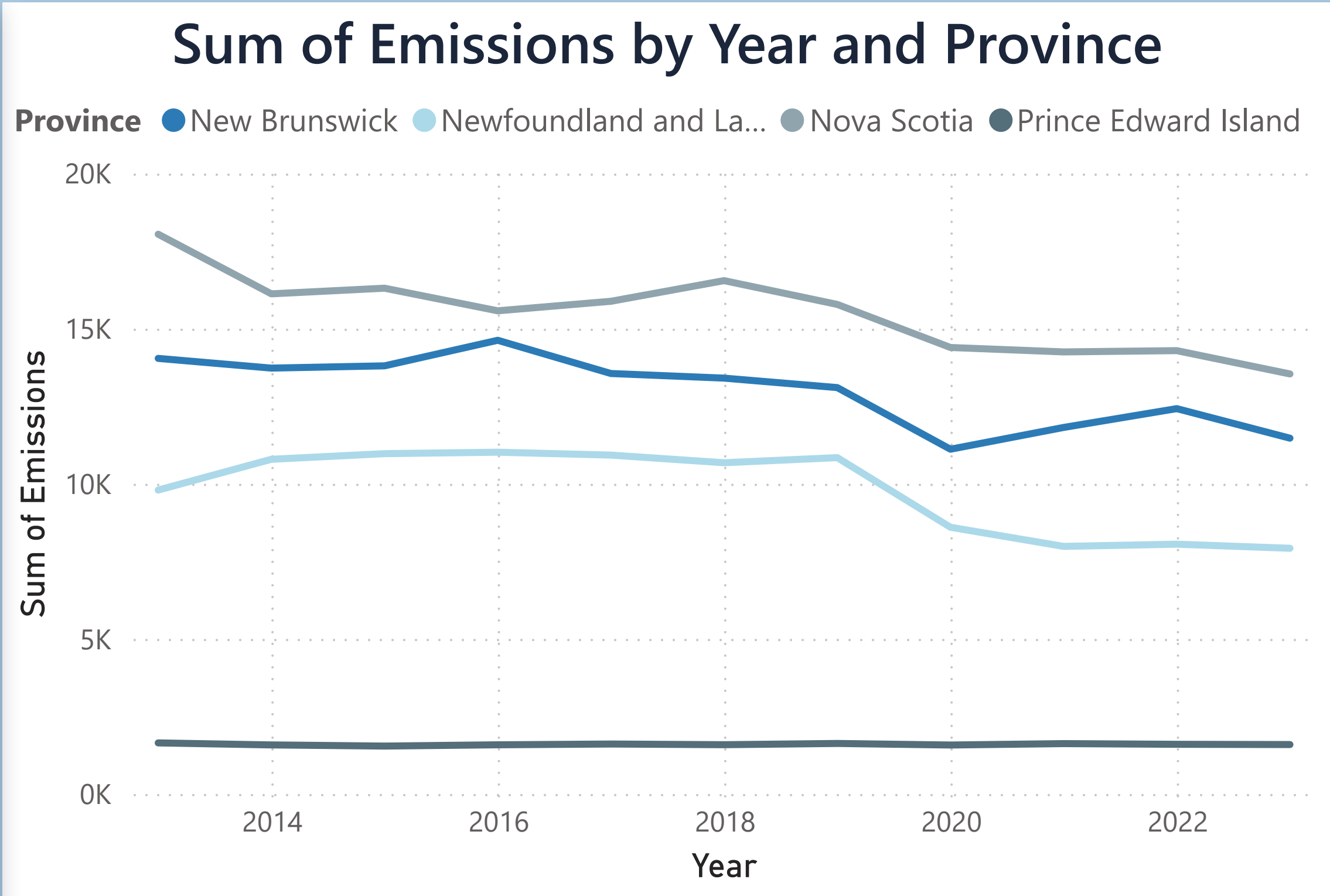
2019

2020

2021

2022

2023



Prov Highest Rank

6

GHG Prov Change %

-20.69%

Total Emissions Prov

438.902K

Total Emissions Prov

4

NS Total Emissions

170

NS GHG Change %

Province Slcr

Alberta	Manitoba	Newfoundland and Labrador	Nova Scotia	Ontario	Quebec	Yukon
British Columbia	New Brunswick	Northwest Territories	Nunavut	Prince Edward Island	Saskatchewan	

Sector Slcr

☐ Accommodation and food services

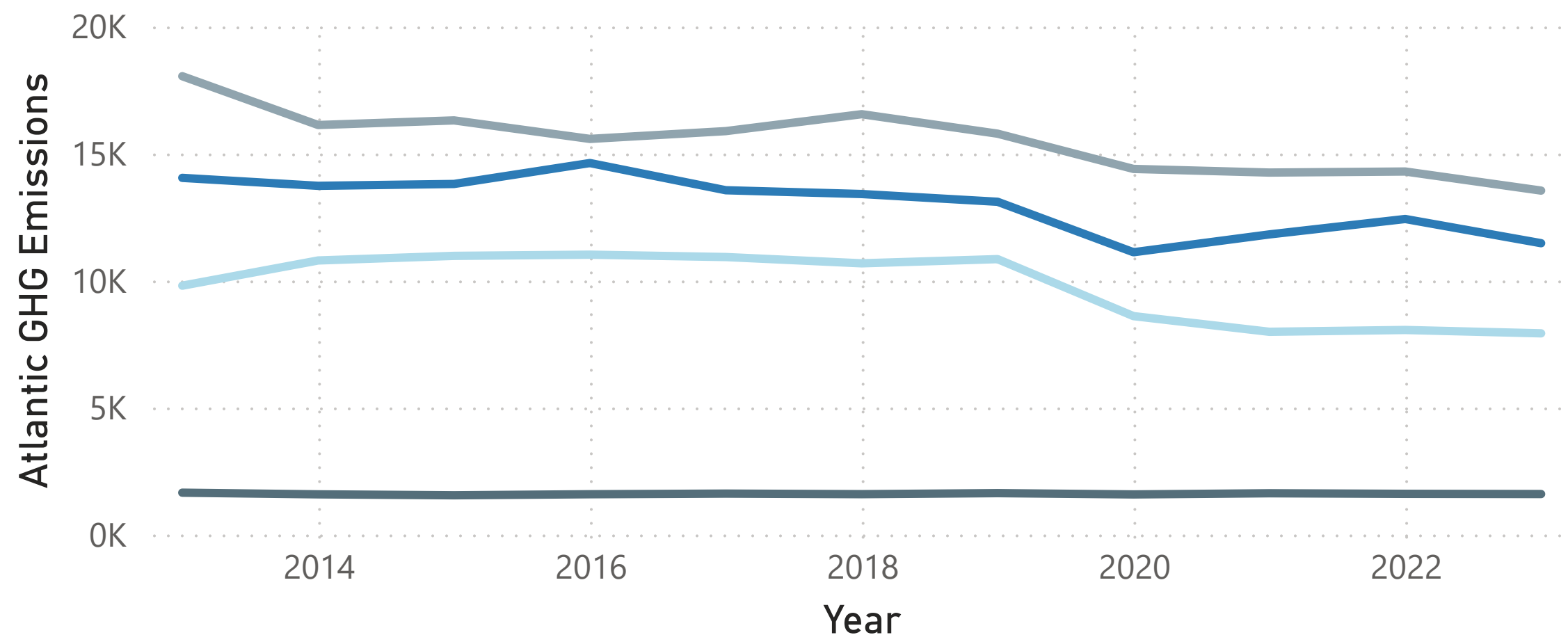
☐ Administrative and support services

☐ Advertising, promotion, meals, entertainment, and recreation

☐ Advertising, public relations and related services

ATLANTIC GHG EMISSIONS TREND BY PROVINCE

Province ● New Brunswick ● Newfoundland and Labr... ● Nova Scotia ● Prince Edward Island



ENVIRONMENTAL TRENDS

Atlantic GHG Emissions

438.902K

Atlantic GHG Change %

-20.69%

NS GHG Change %

-24.96%

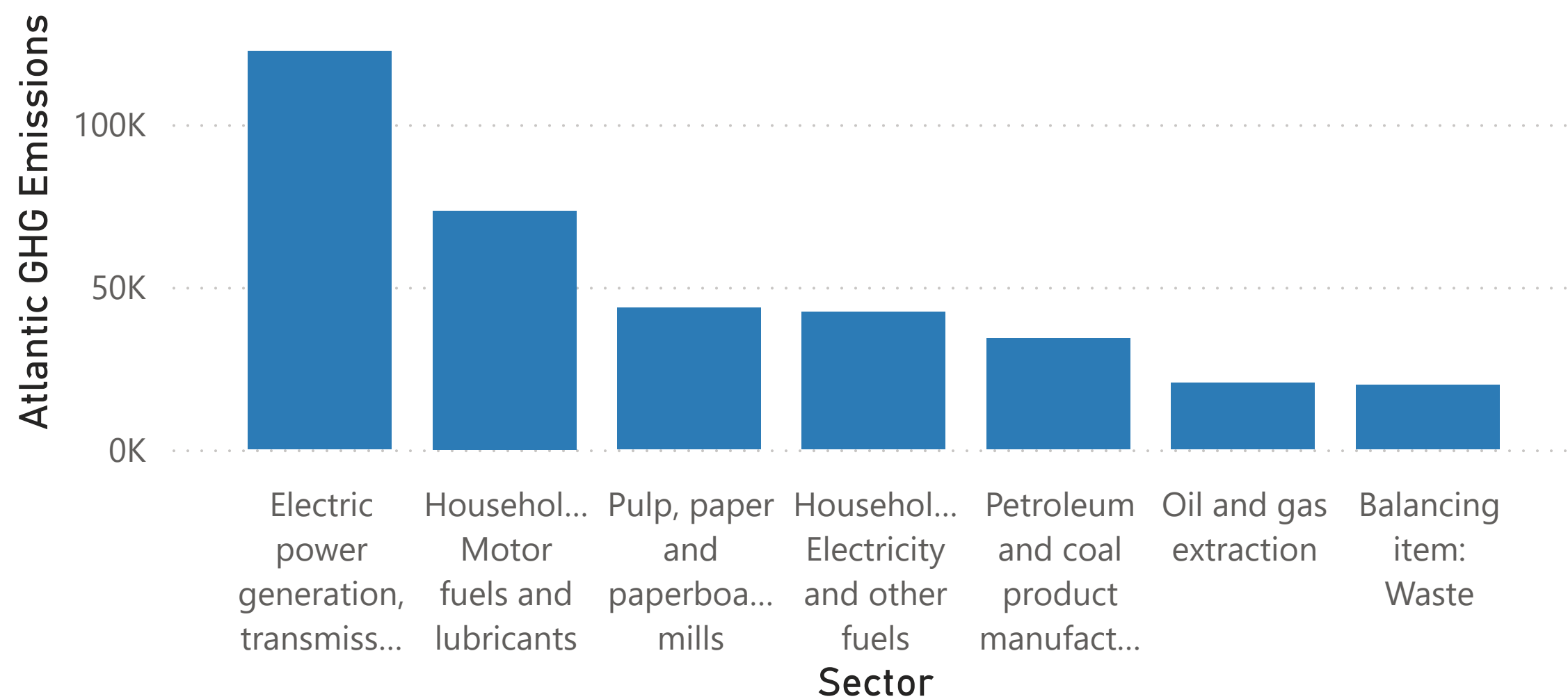
Province Slcr

All

Year Slcr

All

TOP 7 ATLANTIC EMISSION SECTORS



TOTAL EMISSIONS BY ATLANTIC PROVINCE

